

Even Yaru

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ABSTRACT

Rock-Paper-Scissors provides a controlled setting for studying decision-making heuristics under a unique mixed-strategy Nash equilibrium. We ask whether players exhibit the classic Win-Stay/Lose-Shift heuristic, and how this tendency is distributed across a population of 72 players (WXZ2014 dataset, 300 rounds each). A population-level chi-square test reveals only a weak association between action persistence and previous-round outcome (Cramér’s $V = 0.138$), but a player-level analysis reveals substantially stronger individual dependence (mean $V=0.234$, right-skewed), suggesting the pooled statistic masks meaningful heterogeneity. We formalize this heterogeneity using an Emissions-Only GLM-HMM, identifying four latent behavioral archetypes. Surprisingly, one prominent archetype exhibits the reverse pattern — Win-Shift/Lose-Stay — rather than the classic heuristic. An independent hierarchical Bayesian model with player-specific random slopes supports this heterogeneity, and per-player loss-sensitivity estimates from the two architecturally unrelated models correlate strongly ($r = 0.931$, permutation $p < 0.0001$, $n = 10,000$). These convergent results indicate that Win-Stay/Lose-Shift is not a population-uniform strategy, and that a substantial subset of players instead follows its inverse. Code, data and reproduction instructions available at [Code](#), data and reproduction instructions available at [Code](#)

1 Introduction

Rock-Paper-Scissors is a useful testbed for studying feedback-driven decision heuristics precisely because it has a unique, fully-mixed Nash equilibrium (1/3 per action) with no asymmetry between choices. This provides a sharp theoretical benchmark: any systematic dependence between a player’s action and the outcome of the previous round constitutes a clear deviation from optimal play. The classic Win-Stay/Lose-Shift (WSLS) heuristic — repeating a successful action and

abandoning an unsuccessful one — is one such deviation, and has been documented at the population level in prior work on this dataset (Wang et al., 2014).

Existing analyses of WSLS typically report a single population-wide estimate, implicitly assuming the heuristic operates uniformly across individuals. This leaves open a more basic question: is WSLS a population-uniform strategy, or does it vary substantially across players — and if so, does a meaningful fraction of players follow it, follow its reverse, or exhibit no outcome-dependence at all? We address this using three independent approaches: a player-level statistical test of outcome-dependence, an unsupervised hidden-state model (GLM-HMM) that recovers discrete behavioral archetypes without assuming their existence a priori, and a hierarchical Bayesian model that estimates player-specific reactivity directly. Convergence across these architecturally unrelated methods provides robust evidence for the structure we report.

2 Methods

2.1 The Data

We analyze the WXZ2014 dataset (Wang et al., 2014), comprising 72 participants playing 300 rounds of iterated Rock-Paper-Scissors under a random pairwise-matching protocol. After constructing lagged variables (previous choice, previous outcome), 298 rounds per player remain, yielding $N \approx 21,400$ player-round observations used throughout this work.

2.2 Preliminary Statistical Tests

2.2.1 Goodness of Fitting Test

Prior to model fitting, we test two properties of the raw data. First, a goodness-of-fit test evaluates whether choice frequencies match the Nash-equilibrium prediction ($H_0 : p_{\text{Paper}} = p_{\text{Rock}} = p_{\text{Scissors}} = \frac{1}{3}$), using the statistic

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

Where O_i are the observed samples and E_i are expected samples from H_0 . Under H_0 this statistic follows a χ^2 distribution with 2 degrees of freedom. H_0 is rejected if $\chi^2 > \chi_{0.05, (2)}^2 = 5.99$.

2.2.2 χ^2 test of independence and Cramér's V

A 2×3 chi square test of independence evaluates whether Stay/Shift behavior depends on the previous round's outcome

$$\chi^2 = \sum_{r=1}^2 \sum_{c=1}^3 \frac{(O_{r,c} - E_{r,c})^2}{E_{r,c}}$$

Under H_0 , that the samples are independent, this statistic follows a χ^2 distribution with 2 degrees of freedom. H_0 is rejected if $\chi^2 > \chi_{0.05, (2)}^2 = 5.99$.

Rejection of H_0 motivates the input-driven architecture adopted later in Section 2.4. Given the large sample size, significance alone is uninformative, so we also report Cramér's V:

$$V = \sqrt{\frac{\chi^2}{N \cdot (\min(r, c) - 1)}}$$

V rescales χ^2 by sample size (and table dimensionality), yielding a bounded measure of association strength independent of N. The closer the V is to 1, the stronger the dependence, and the closer to 0, the weaker.

A single population-wide V may mask heterogeneous player-level relationships, so we additionally compute V separately for each player's own 2×3 table and report the resulting distribution.

2.3 Classical Hidden Markov Model (HMM)

A Hidden Markov Model (HMM) is a doubly stochastic probabilistic framework that assumes an observed sequence $Y = (y_1, y_2, \dots, y_t)$ is directly generated by an underlying, unobserved (latent) Markov process with a finite state space $S \in \{1, 2, \dots, K\}$.

A standard HMM is defined by three fundamental components:

1. **Initial State Distribution (π):** The probability distribution over the latent states at the initial time step $t = 1$.
2. **State Transition Matrix (A):** The probabilities governing transitions between latent states, following the first-order Markov property where S_t depends only on S_{t-1} . In a standard HMM, these transitions are stationary and time-invariant.
3. **Emission Distribution (B):** The conditional probability distribution connecting the hidden state to the observed output, specifying $P(y_t | S_t = k)$ —the likelihood of observing a specific action given that the system is currently in latent state k .
Specifically, in our case where actions are binary ($y_t = 1$ for staying with the same choice as the previous round, and $y_t = 0$ for changing it), the emission follows a Bernoulli distribution, parametrized by: $\theta_k = P(\text{Stay} | S_t = k)$, such that:

$$P(y_t | S_t = k) = \theta_k^{y_t} \cdot (1 - \theta_k)^{1-y_t}$$

2.3.1 Motivation for the GLM-HMM

The standard Hidden Markov Model (HMM) suffers from a key limitation that prevents it from fully capturing the behavioral dynamics in our experiment:

Exogenous Triggers (Lack of Feedback Integration) In a classic HMM, the observed action at trial t ($y_t \in \{\text{Stay}, \text{Shift}\}$) depends only on the current hidden state (S_t). This structure ignores the outcome of the previous trial ($X_t = [X_t^{\text{Win}}, X_t^{\text{Tie}}, X_t^{\text{Lose}}] \in \mathbb{R}^3$) and thus cannot evaluate how immediate win or loss feedback directly influences a player's tendency stay or shift his decision, given the current hidden state ($P(y_t = 1 | S_t = k)$).

2.4 Emissions GLM-HMM

To address this limitation, we extend the classical HMM by making the emission distribution input-driven, while keeping the transition matrix A homogeneous (i.e., time-invariant, as in the classical HMM of Section 2.3). The probability of staying with the previous choice ($y_t = 1$) given the current state and the outcome of the previous

round is modeled via state-dependent logistic regression:

$$P(y_t = 1 | S_t = k, X_t) = \sigma(w_{k,0} + W_k \cdot X_t)$$

where: $\sigma := \frac{1}{1+e^{-x}}$ is the Sigmoid function, $w_{k,0}$ is the state-specific baseline intercept, and W_k is the weight vector capturing the behavioral influence of feedback X_t within state k .

The model's goal is to estimate the parameters

$$\Theta = \left\{ \pi, A, \{w_{k,0}, W_k\}_{k=1}^K \right\}$$

The number of hidden states K is selected by comparing AIC/BIC across candidate values (Section 3.2.2).

$w_{k,0}$ are the baseline intercept, representing the tendency to stay with the same choice in state k regardless of the feedback.

W_k are the slope weights — a weight vector measuring the directional influence of feedback X_t on the emission probability within state k .

The estimation of the parameters is done via the **EM** algorithm to maximize the marginal Log-Likelihood $\log P(Y|X)$:

2.4.1 The E-step

In this step we calculate the expectation of the Complete-Data-Log-Likelihood with respect to the posterior distribution of the hidden states, based on the previous iteration $\Theta^{(\text{old})}$:

$$Q(\Theta; \Theta^{(\text{old})}) = \mathbb{E}_{S|Y, X; \Theta^{(\text{old})}} [\log P(Y, S|X; \Theta)]$$

The calculation is done by the Forward-Backward method to obtain the posterior responsibilities:

$\gamma_{t,k} := P(S_t = k | Y, X; \Theta^{(\text{old})})$ which is the probability of being in state k at time t given the entire observed sequence (Y) .

$\xi_{t,i,j} := P(S_{t-1} = i, S_t = j | Y, X; \Theta^{(\text{old})})$ which is the transition probability from state i to state j at time t given the entire observed sequence (Y) .

Setting these responsibilities into $Q(\Theta; \Theta^{(\text{old})})$, we get:

$$Q(\Theta; \Theta^{(\text{old})}) = \sum_{k=1}^K \gamma_{1,k} \log \pi_k +$$

$$\sum_{t=1}^T \sum_{k=1}^K \gamma_{t,k} \log P(Y_t | S_t = k; w_{k,0}, W_k) +$$

$$\sum_{t=2}^T \sum_{i=1}^K \sum_{j=1}^K \xi_{t,i,j} \log A_{i,j}$$

2.4.2 The M-step

In the M-step we want to maximize $Q(\Theta; \Theta^{(\text{old})})$. Because the Log-Likelihood function is additively separable in its parameters, we can optimize each group of parameters separately.

1. By $w_{k,0}, W_k$:

$$Q_{\text{emission}}(w, W) =$$

$$\sum_{t=1}^T \sum_{k=1}^K \gamma_{t,k} \log P(y_t | S_t = k, X_t; w_{k,0}, W_k)$$

Notice that this is exactly the Weighted GLM, where $\gamma_{t,k}$ act as the sample weights. This expression doesn't have a closed form solution. Thus the parameters $w_{k,0}$ and W_k will be updated iteratively using gradient-based numerical optimization algorithms (e.g., L-BFGS or Newton-Raphson).

2. By A :

To maximize $\sum_{t,i,j} \xi_{t,i,j} \log A_{i,j}$ under the constraint $\sum_j A_{i,j} = 1$, we use Lagrange multipliers, and get the closed form:

$$A_{i,j} = \frac{\sum_{t=2}^T \xi_{t,i,j}}{\sum_{t=2}^T \sum_{j'=1}^K \xi_{t,i,j'}}$$

2.4.3 Bootstrap

Because trial-level observations are serially dependent, the i.i.d. assumption underlying the standard bootstrap is violated at the trial level. We therefore bootstrap at the player level: each of $B = 35$ iterations resamples 72 players with replacement, preserving each player's full temporal sequence, and refits the model. Confidence intervals use the percentile method; because EM state labels are arbitrary up to permutation, each replicate is matched to a canonical state ordering before aggregation. Given the 12 simultaneous

comparisons (4 states $\times$ 3 outcomes), we apply a Bonferroni correction ($\alpha = \frac{0.05}{12} \approx 0.0042$).

2.5 Hierarchical Bayesian Model

To directly capture individual differences between players, we build a hierarchical logistic regression that predicts the probability of staying with the previous choice, based on the outcome of the previous round (Win, Lose, or Tie). Since these three outcomes are mutually exclusive and cover all possibilities, each outcome gets its own coefficient, and no separate intercept is included (adding one would make the model unidentifiable, since the three outcome indicators always sum to 1).

For each player i , we define:

$$\begin{aligned} \text{logit}(P(\text{Stay}_t = 1)) = & \beta_{\text{win},i} \cdot \text{win}_t + \\ & + \beta_{\text{lose},i} \cdot \text{lose}_t + \beta_{\text{tie},i} \cdot \text{tie}_t + \\ & + \beta_{\text{period}} \cdot \text{period}_t + \beta_{\text{lag1}} \cdot \text{stay}_{t-1} \end{aligned}$$

Each of $\beta_{\text{win},i}, \beta_{\text{lose},i}, \beta_{\text{tie},i}$ is drawn from a shared population-level normal distribution (for example, $\beta_{\text{lose},i} = \mu_{\beta_{\text{lose}}} + \sigma_{\beta_{\text{lose}}} \cdot z_i$ with $z_i \sim N(0, 1)$). This gives every player their own sensitivity to each outcome, while still sharing information across all 72 players: a player with little data is pulled closer to the population average, while a player with more data is estimated mostly from their own choices. The population mean μ_{β} tells us the average effect across all players, while the population standard deviation σ_{β} tells us how much players actually differ from each other.



Figure 1: Diagram of the Hierarchical Bayesian Model

Because each player’s coefficients are drawn from a shared population distribution rather than estimated independently, information is shared across players — a property known as partial pooling. In practice, this means a player with many observed losses will have their $\beta_{\text{loss},i}$ estimated mostly from their own data, while a player with very few losses will have their estimate pulled (“shrunk”) toward the population mean $\mu_{\beta_{\text{loss}}}$, since there is not enough individual evi-

dence to override the population-level pattern. This lies between two extremes: a single shared coefficient for all players (which ignores individual differences entirely), and a fully separate regression per player (which would produce unreliable estimates for players with limited data).

We use weakly informative priors on the population-level parameters ($\mu_{\beta} \sim N(0, 1.5)$, $\sigma_{\beta} \sim \text{HalfNormal}(1)$) and sample the posterior using the No-U-Turn Sampler (NUTS), through the numpyro backend, with 1000 tuning steps and 1000 draws.

3 Results

3.1 Preliminary Statistical Tests

3.1.1 Goodness of fitting test

The statistic value computed is $\chi^2 = 48.984$, and the p-value is $p < 10^{-7}$, thus we reject H_0 and the choice frequencies significantly deviate from the Nash-equilibrium.

3.1.2 χ^2 test of independence and Cramér’s V

The computed statistic is $\chi^2 = 407.018$, yielding a p-value of $p < 10^{-89}$. The computed pooled Cramér’s V is 0.138, indicating a weak dependency between samples, but looking at the histogram of Cramér’s V computed player-wise, a more nuanced picture emerges:

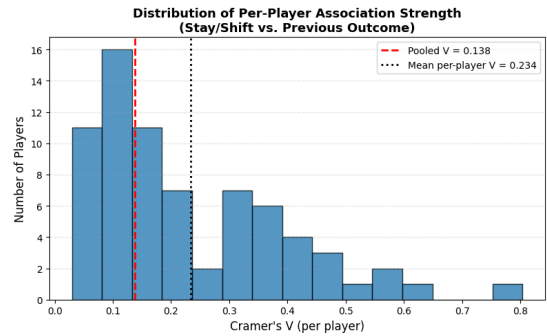


Figure 2: Cramér’s V per-player

As shown in Figure 2, the per-player distribution is right-skewed: most players show weak outcome-dependence, while a smaller subset shows moderate-to-strong dependence, extending the distribution’s tail up to $V \approx 0.8$. This heterogeneity is examined further via the hidden-state models below.

3.2 HMM

3.2.1 Choosing a model

To establish the optimal cognitive architecture for modeling player strategy switching in Rock-Paper-Scissors, we systematically compared the classic HMM model and three candidate GLM-HMM formulations:

1. Transitions-Only (Input-Driven HMM): Feedback dynamics drive state transitions, while choice emission within states remains unconditioned.
2. Emissions-Only GLM-HMM: Choice probabilities within states are dynamically modulated by feedback history (X_t), whereas state transitions follow a static Markovian structure.
3. Full GLM-HMM (Combined): Feedback drives both hidden-state transitions and intra-state choice selection.

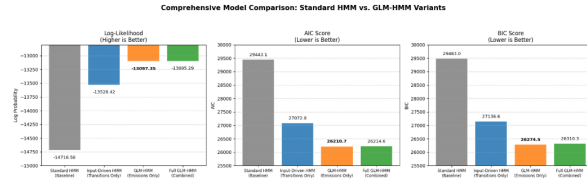


Figure 3: Models GOF

First, we can see that the worst model is the classic HMM, as we predicted in Section 2.3.1, it underfits the data.

Empirical comparison confirms the clear superiority of the Emissions-Only architecture (Figure 3) :

Emissions vs. Transitions: Dynamic choice emission yields a massive fit improvement over transition-only feedback (Δ AIC = 862.1, Log-Likelihood rose from $-13,528.42$ to $-13,097.35$), proving feedback primarily drives action selection rather than state shifts.

Emissions vs. Full Model: The Full model offers a negligible Log-Likelihood gain ($-13,095.29$ vs. $-13,097.35$, Δ LL = 2.06) despite 6 additional parameters. AIC and BIC criteria penalize this extra complexity, favoring Emissions-Only in both AIC (26,210.7 vs. 26,214.6) and BIC (26,274.5 vs. 26,310.3, Δ BIC = 35.8).

Consequently, input-driven transition dynamics offer no substantial improvement over the stan-

dard GLM-HMM, allowing us to proceed strictly with the emissions-only model.

3.2.2 Choosing K

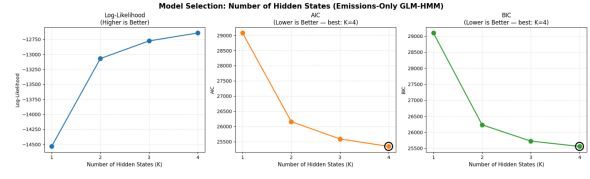


Figure 4: Choosing K

In order to choose the number of hidden states K , we compared the GOF metrics (Log-Likelihood, AIC, BIC) for $K = 1, \dots, 4$. As shown in Figure 4, model fit improves monotonically as K increases, with no sign of a plateau in the range tested — that is, from a purely statistical standpoint, there is no guarantee that higher values of K would not yield an even better fit. We chose to stop at $K = 4$ for reasons of interpretability: This number is sufficient to identify four distinct behavioral patterns with clear psychological meaning (see Section 3.2.3), whereas adding further hidden states would compromise our ability to interpret each state as a distinct behavioral archetype. We therefore treat $K=4$ as a practical ceiling, rather than necessarily the model with the lowest achievable BIC.

3.2.3 Emission Parameter Analysis

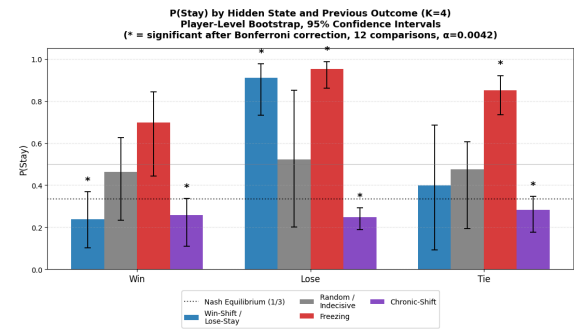


Figure 5: Emission Probabilities $P(\text{Stay})$ by Hidden State and Previous Outcome, with 95% Bootstrap CIs

To characterize the decision-making rules operating within each hidden state, we examined the state specific stay probabilities $P(\text{Stay})$ corresponding to feedback history (Win, Lose, Tie). Given the 12 simultaneous comparisons ($4 \text{ states} \times 3 \text{ outcomes}$), we apply a Bonferroni correction.

* in Figure 5 denote coefficients remaining significant after this correction.

Behavioral Interpretation

1. Win-Shift / Lose-Stay (The blue state)

This state demonstrates significant asymmetric behavior. After a loss, players show a clear tendency to stay with the same choice ($P(\text{Stay}) = 0.91$), with the CI entirely above $p = 0.5$. Conversely, after a win, players show the opposite tendency — they shift their choice with high probability ($P(\text{Stay}) = 0.238$), and this tendency is also significant. After a tie, the response is not significant.

2. Chronic-Shift (The purple state)

This state exhibits the most consistent and significant pattern across all three outcomes. Regardless of the outcome of the previous round, the CI of $P(\text{Stay})$ remains below $p = 0.5$. Moreover, after a loss, even the upper bound of the CI falls below the Nash equilibrium value ($p = \frac{1}{3}$).

3. Freezing (The red state)

This state represents consistency and persistence. After a tie or a loss, players stay with the same choice with high probability, with CIs entirely above $p = 0.5$. After a win, there is still a tendency to stay, but the CI crosses the $p = 0.5$ bar — while remaining entirely above the Nash equilibrium. This behavior is therefore significant with respect to the Nash equilibrium, but not with respect to binary indifference.

4. Random / Indecisive (The gray state)

This state represents a complete absence of a clear strategy. Means across all three previous outcomes are close to 0.5, with CIs crossing both benchmarks.

3.2.4 Analysis of a Single Game

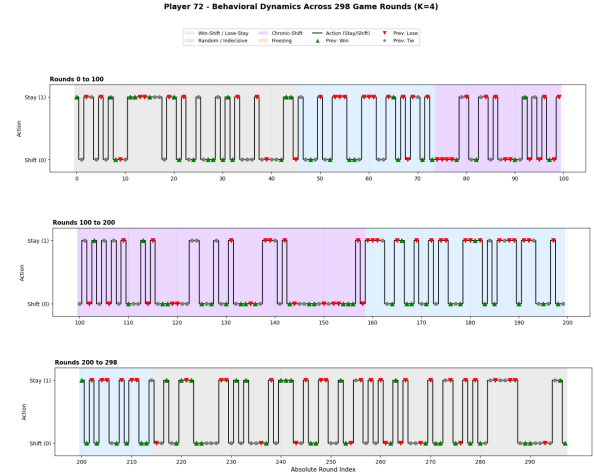


Figure 6: Behavioral Dynamics. **Watch the full animation by clicking [HERE](#)¹**

Consider the game trajectory of player 72, shown in Figure 6. The figure displays all of this player's rounds and outcomes, with the background indicating the hidden state the player was in, along with whether they stayed with or shifted away from their previous choice.

For this player, the model classifies them as Random up to round 45 — a regime in which it is not possible to predict whether they will stay with their choice or not. From round 45 to round 75, they enter the Win-Shift / Lose-Stay state. This transition can be seen directly by eye: after a loss, the player tends to stay with their choice, while after a win, they tend to shift.

From round 75 to round 168, the player enters the Chronic-Shift state. Visually, one can see that for most of this period the player shifts their choice regardless of outcome.

The player then re-enters the Win-Shift / Lose-Stay state until round 215, after which the model classifies them as Random for the remainder of the game.

¹The algorithm used to compute the real-time probability that the player is in a given hidden state is not presented here due to space constraints.

3.2.5 Occupancy Over Time Analysis

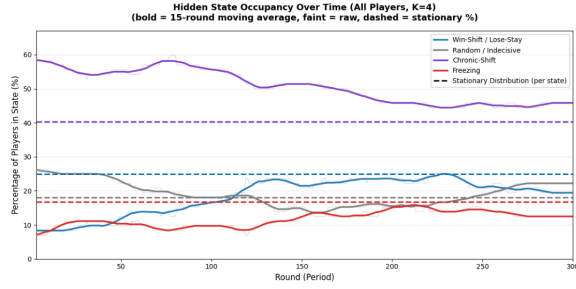


Figure 7: Occupancy over time

Figure 7 reveals two key patterns. First, by the stationary distribution, Chronic-Shift is the most dominant state ($\sim 40\%$), followed by Win-Shift/Lose-Stay ($\sim 25\%$), with Random and Freezing trailing closely ($\sim 17 - 18\%$ each). Second, there is a clear temporal dynamic: Chronic-Shift dominates early in the game (over 50% of players), gradually declining across rounds, while Win-Shift/Lose-Stay — nearly the rarest state at the outset — becomes increasingly central as the game progresses.

3.3 Hierarchical Bayesian Model

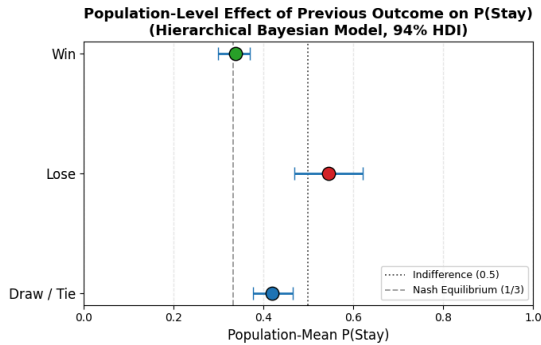


Figure 8: Bayesian Coefficients

Examining the population-mean effect in Figure 8 against two relevant benchmarks — binary indifference ($p = 0.5$) and the Nash-equilibrium baseline ($p = \frac{1}{3}$) — a pattern emerges. After a win, the mean response ($P(\text{Stay}) = 0.337$) is statistically indistinguishable from Nash play (CI includes $\frac{1}{3}$), while remaining significantly below indifference; players' choices following a win are consistent with unbiased random play rather than any systematic persistence or avoidance. After a Draw, the response (0.420) is significant relative to both benchmarks: below indifference, yet still above the Nash baseline. After a Lose, the pattern reverses: the mean (0.546) is significant relative to

Nash — players persist more than chance would predict — but not relative to indifference, and the wide CI reflects substantial between-player disagreement, consistent with the large $\sigma_{\beta_{\text{lose}}}$ reported below.

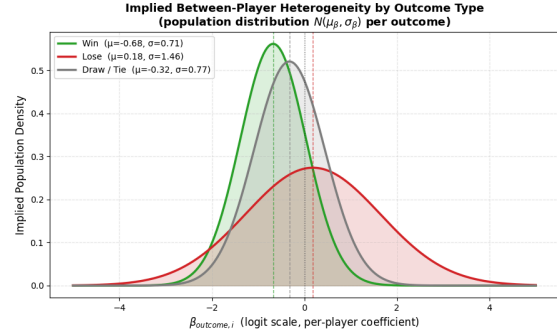


Figure 9: Implied Between-Player Heterogeneity by Outcome Type

Figure 9 illustrates the heterogeneity implied by the hierarchical model: the implied distribution of $\beta_{\text{lose},i}$ is substantially wider ($\sigma = 1.46$) than $\beta_{\text{win},i}$ ($\sigma = 0.71$) and $\beta_{\text{tie},i}$ ($\sigma = 0.77$) indicating greater between-player variability in response to losing than to winning or tying.

3.4 Comparing the models

Having established each model independently, we next assess whether the two frameworks agree on the structure of the underlying behavior, using two complementary criteria: predictive accuracy on unseen players, and direct correlation of player-level estimates.

3.4.1 Out-of-Sample Predictive Performance

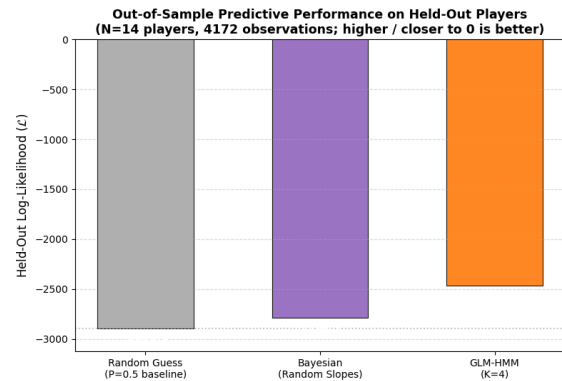


Figure 10: Out-of-Sample Predictive Performance on Held-Out Players

To evaluate predictive performance, players (not rounds) were split 80/20 into train/test sets, with test players entirely unseen by either

model. Bayesian held-out log-likelihood uses the log of the posterior-mean predictive probability, avoiding downward bias from unconstrained per-player effects. Agreement between models is assessed via a permutation test (10,000 resamples) correlating each player’s Bayesian $\beta_{\text{lose},i}$ with an occupancy-weighted GLM-HMM score, $\sum_k o_{i,k} \beta_{\text{lose},k}$.

Held-out log-likelihood was -2788.22 for the hierarchical Bayesian model and -2468.03 for the Emissions-Only GLM-HMM as shown in Figure 10, indicating that the GLM-HMM model achieves better out-of-sample predictive performance. Furthermore, for a naive random guess ($p = 0.5$), the LL is -2891.81 . Both models therefore outperform random guessing.

3.4.2 Convergent Validity of Player-Level Estimates

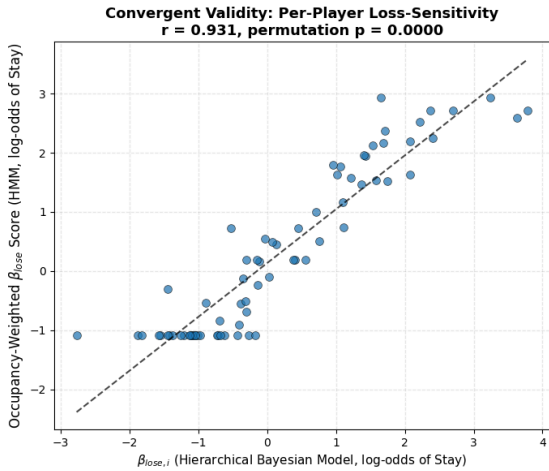


Figure 11: Convergent Validity

For each player, we compared their loss-sensitivity coefficient from the hierarchical Bayesian model ($\beta_{\text{lose},i}$) to an occupancy-weighted loss-sensitivity score derived independently from the GLM-HMM. The two measures were strongly correlated (Pearson $r = 0.931$, permutation $p < 0.0001$, $n = 10,000$; Figure 11), despite arising from architecturally unrelated models.

4 Discussion

Three architecturally independent methods—a player-level χ^2 analysis, an unsupervised GLM-HMM, and a hierarchical Bayesian model with player-specific coefficients—converge on the same conclusion: Win-Stay/Lose-Shift is

not a population-uniform heuristic. The pooled Cramér’s V (0.138) substantially understates the true strength of outcome-dependence for a meaningful subset of players (mean per-player $V = 0.234$), and the GLM-HMM resolves this heterogeneity into four discrete archetypes. Critically, the Bayesian model—built on a continuous, architecturally unrelated framework—independently recovers the same individual differences ($r = 0.931$, permutation $p < 0.0001$), providing strong convergent validity rather than an artifact of any single modeling choice. Notably, the second most prevalent archetype exhibits the reverse of the classic heuristic—shifting after wins and staying after losses—a pattern consistent with sophisticated (level-2) opponent-modeling rather than noise.

Several limitations warrant mention. First, while we tested whether feedback-driven transition dynamics improve model fit (Section 3.2.1) and found no improvement over the time-homogeneous Emissions-Only architecture, we did not test whether transitions depend on round number; Bayesian model detects a modest population-level increase in overall persistence across rounds ($\beta_{\text{period}} = 0.066$, 95% CI [0.039, 0.094]), suggesting session-length effects that a purely feedback-driven architecture cannot capture.

Second, collapsing choices into a binary Stay/Shift outcome discards directional information (e.g., Rock→Paper vs. Rock→Scissors); prior work on this dataset shows these directions are not symmetric (Xu et al., 2013), so archetypes identified here may themselves be directionally heterogeneous.

Third, out-of-sample prediction for unseen players remained modest for both models, indicating that predicting a new individual from group-level parameters alone remains difficult.

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